1.4 Complex Numbers

The history of mathematics shows that man has been developing and enlarging his concept of **number** according to the saying that "Necessity is the mother of invention". In the remote past they started with the set of counting numbers and invented, by stages, the negative numbers, rational numbers irrational numbers. Since square of a positive as well as negative number is a positive number, the square root of a negative number does not exist in the realm of real numbers. Therefore, square roots of negative numbers were given no attention for centuries together. However, recently, properties of numbers involving square roots of negative numbers have also been discussed in detail and such numbers have been found useful and have been applied in many branches of pure and applied mathematics. The numbers of the form $x+iy$, where $x, y \in \mathbb{R}$ and $i=\sqrt{-1}$, are called **complex numbers,** here x is called **real part** and y is called **imaginary part** of the complex number. For example, $3+4i$, $2-\frac{5}{7}i$ etc. are complex numbers.

Note: Every real number is a complex number with 0 as its imaginary part.

Let us start with considering the equation.

$$x^2+1=0 \qquad (1)$$

$$\Rightarrow \quad x^2=-1$$

$$\Rightarrow \quad x=\pm\sqrt{-1}$$

$\sqrt{-1}$ does not belong to the set of real numbers. We, therefore, for convenience call it **imaginary number** and denote it by i (read as iota).

The product of a real number and i is also an **imaginary number**. Thus $2i, -3i, \sqrt{5}i, -\frac{11}{2}i$ are all imaginary numbers. i which may be written $1.i$ is also an imaginary number.

Powers of i :

$$i^2=-1 \quad \text{(by definition)}$$

$$i^3=i^2.i=-1.i=-i$$

$$i^4=i^2\times i^2=(-1)(-1)=1$$

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Thus any power of i must be equal to $1, i, -1$ or $-i$. For instance,

$$i^{13} = (i^2)^6 . i = (-1)^6 . i = i$$

$$i^6 = (i^2)^3 = (-1)^3 = -1 \quad \text{etc.}$$

1.4.1 Operations on Complex Numbers

With a view to develop algebra of **complex numbers**, we state a few definitions. The symbols a,b,c,d,k, where used, represent real numbers.

1) $a+bi=c+di \Rightarrow a=c \wedge b=d.$

2) Addition: $(a+bi)+(c+di)=(a+c)+(b+d)i$

3) $k(a+bi)=ka+kbi$

4) $(a+bi)-(c+di) = (a+bi)+[-(c+di)]$

$$= a+bi+(-c-d\,i)$$

$$= (a-c)+(b-d)i$$

5) $(a+bi).(c+di)=ac+adi+bci+bd\,i^2 = (ac-bd)+(ad+bc)i.$

6) **Conjugate Complex Numbers:** *Complex numbers* of the form $(a+bi)$ and $(a-bi)$ which have the same real parts and whose imaginary parts differ in sign only, are called conjugates of each other. Thus $5+4\,i$ and $5-4\,i$, $-2+3i$ and $-2-3\,i$, $-\sqrt{5}\,i$ and $\sqrt{5}\,i$ are three pairs of conjugate numbers.

Note: A real number is self-conjugate.

1.4.2 Complex Numbers as Ordered Pairs of Real Numbers

We can define complex numbers also by using ordered pairs.

Let C be the set of ordered pairs belonging to $\mathbb{R} \times \mathbb{R}$ which are subject to the following properties: -

i) $(a,b)=(c,d) \Leftrightarrow a=c \wedge b=d.$

ii) $(a,b)+(c,d)=(a+c,b+d)$

iii) If k is any real number, then $k(a,b)=(ka,kb)$

iv) $(a,b)\,(c,d) = (ac-bd,\ ad+bc)$

Then C is called the set of *complex numbers*. It is easy to see that

$(a,b)-(c,d) = (a-c,\ b-d)$

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Properties (1), (2) and (4) respectively define equality, sum and product of two complex numbers. Property (3) defines the product of a real number and a complex number.

Example 1: Find the sum, difference and product of the complex numbers (8,9) and (5,–6)

Solution: Sum = (8+5, 9–6) = (13,3)

Difference = (8 – 5, 9 – (–6)) = (3,15)

Product = (8.5 – (9) (–6), 9.5 + (–6) 8)

= (40 + 54, 45 – 48)

= (94, –3)

1.4.3 Properties of the Fundamental Operations on Complex Numbers

It can be easily verified that the set C satisfies all the field axioms i.e., it possesses the properties 1(i to v), 2 (vi to x) and 3(xi) of Art. 1.3.

By way of explanation of some points we observe as follows: -

i) The additive identity in C is (0, 0).

ii) Every complex number (a, b) has the additive inverse $(-a, -b)$ i.e., $(a, b) + (-a, -b) = (0,0)$.

iii) The multiplicative identity is (1,0) i.e.,

$(a,b).(1,0) = (a.1 - b.0,\ b.1 + a.0) = (a, b).$

$=(1,0)\ (a, b)$

iv) Every non-zero complex number {i.e., number not equal to (0,0)} has a multiplicative inverse.

The multiplicative inverse of (a, b) is $\left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$

$$(a, b)\left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right) = (1,0), \text{ the identity element}$$

$$= \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)(a,b)$$

v) $(a, b)\ [(c, d) \pm (e, f)] = (a, b)(c, d) \pm (a, b)(e, f)$

Note: The set C of complex numbers does not satisfy the order axioms. In fact there is no sense in saying that one complex number is greater or less than another.

1.4.4 A Special Subset of C

We consider a subset of C whose elements are of the form $(a, 0)$ i.e., second component of each element is zero.

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Let $(a,0)$, $(c,0)$ be two elements of this subset. Then

i) $(a,0)+(c,0)=(a+c,0)$ ii) $k(a,0)=(ka,0)$

iii) $(a,0)\times(c,0)=(ac,0)$

iv) Multiplicative inverse of $(a,0)$ is $\left(\frac{1}{a},0\right)$, $a\neq 0$.

Notice that the results are the same as we should have obtained if we had operated on the real numbers a and c ignoring the second component of each ordered pair i.e., 0 which has played no part in the above calculations.

On account of this special feature we identify the complex number $(a, 0)$ with the real number a i.e., we postulate:

$$(a,0)=a \qquad (1)$$

Now consider (0, 1)

$(0, 1) . (0,1) = (-1,0)$

$= -1$ (by (1) above).

If we set $(0,1) = i$ (2)

then $(0,1)^2 = (0,1)(0,1) = i.i = i^2 = -1$

We are now in a position to write every complex number given as an ordered pair, in terms of i. For example

$$\begin{aligned}(a,b) &= (a,0)+(0,b) && \text{(def. of addition)}\\ &= a(1,0)+b(0,1) && \text{(by (1) and (2) above)}\\ &= a.1+bi\\ &= a+ib\end{aligned}$$

Thus $(a,b) = a + ib$ where $i^2 = -1$

This result enables us to convert any Complex number given in one notation into the other.

Exercise 1.2

1. Verify the addition properties of complex numbers.
2. Verify the multiplication properties of the complex numbers.
3. Verify the distributive law of complex numbers.

$$(a, b)[(c,d)+(e,f)] = (a,b)\,(c,d)+(a,b)\,(e.f)$$

(**Hint:** Simplify each side separately)

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4. Simplify the following:

i) i^9 ii) i^{14} iii) $(-i)^{19}$ iv) $(-1)^{\frac{-21}{7}}$

5. Write in terms of i

i) $\sqrt{-1}b$ ii) $\sqrt{-5}$ iii) $\sqrt{\frac{-16}{25}}$ iv) $\sqrt{\frac{1}{-4}}$

Simplify the following:

6. $(7, 9) + (3, -5)$ 7. $(8,-5) - (-7, 4)$ 8. $(2,6)\,(3,7)$

9. $(5, -4)\ (-3, -2)$ 10. $(0,3)\,(0,5)$ 11. $(2,6) \div (3,7)$.

12. $(5, -4) \div (-3, -8)$ $\left(\text{Hint for 11: } \frac{(2,6)}{(3,7)} = \frac{2+6i}{3+7i} \times \frac{3-7i}{3-7i} \quad \text{etc.}\right)$

13. Prove that the sum as well as the product of any two conjugate complex numbers is a real number.

14. Find the multiplicative inverse of each of the following numbers:

i) $(-4, 7)$ ii) $\left(\sqrt{2}, -\sqrt{5}\right)$ iii) $(1,0)$

15. Factorize the following:

i) $a^2 + 4b^2$ ii) $9a^2 + 16b^2$ iii) $3x^2 + 3y^2$

16. Separate into real and imaginary parts (write as a simple complex number): -

i) $\frac{2-7i}{4+5i}$ ii) $\frac{(-2+3i)^2}{(1+i)}$ iii) $\frac{i}{1+i}$

1.5 The Real Line

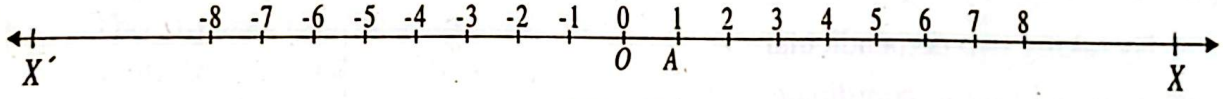


Fig. (1)

In Fig.(1), let $\overleftrightarrow{X'\ X}$ be a line. We represent the number 0 by a point O (called the origin) of the line. Let $|OA|$ represents a unit length. According to this unit, positive numbers are represented on this line by points to the right of O and negative numbers by points to the left of O. It is easy to visualize that all +ve and –ve rational numbers are represented on this line. What about the irrational numbers?

The fact is that all the irrational numbers are also represented by points of the line. Therefore, we postulate: -

Postulate: A (1 – 1) correspondence can be established between the points of a line l and the real numbers in such a way that: -

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i) The number 0 corresponds to a point O of the line.

ii) The number 1 corresponds to a point A of the line.

iii) If x_1, x_2 are the numbers corresponding to two points P_1, P_2, then the distance between P_1 and P_2 will be $|x_1 - x_2|$.

It is evident that the above correspondence will be such that corresponding to any real number there will be one and only one point on the line and vice versa.

When a (1 – 1) correspondence between the points of a line $\overleftrightarrow{x' \quad x}$ and the real numbers has been established in the manner described above, the line is called the **real line** and the real number, say x, corresponding to any point P of the line is called the **coordinate** of the point.

1.5.1 The Real Plane or The Coordinate Plane

We know that the *cartesian product* of two non-empty sets A and B, denoted by $A \times B$, is the set:

$A \times B = \{ (x, y) \mid x \in A \wedge y \in B \}$

The members of a *cartesian* product are **ordered pairs.**

The *cartesian* product $\mathbb{R} \times \mathbb{R}$ where $\mathbb{R}$ is the set of real numbers is called the **cartesian plane.**

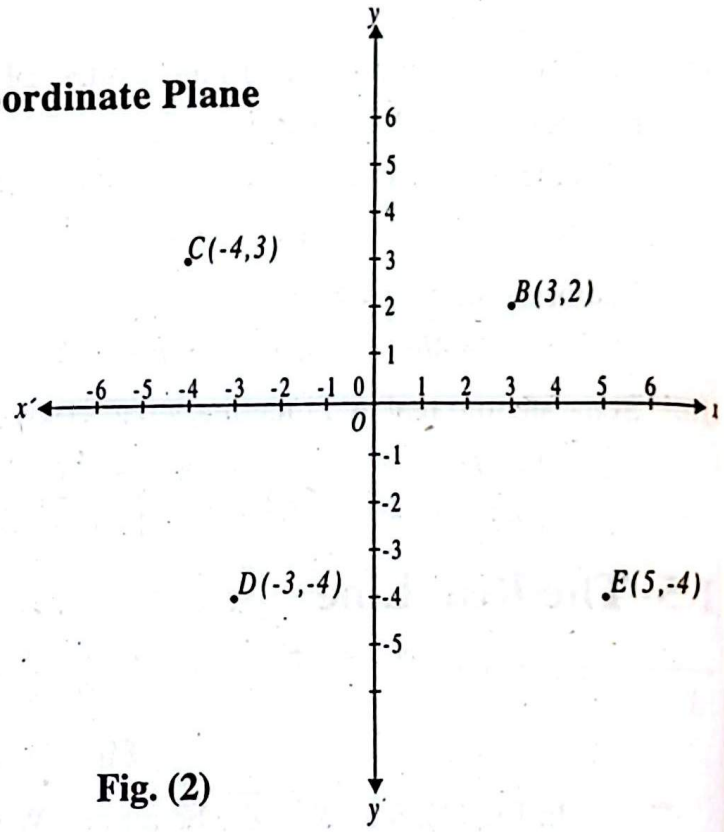


Fig. (2)

By taking two perpendicular lines $x'ox$ and $y'oy$ as coordinate axes on a geometrical plane and choosing a convenient unit of distance, elements of $\mathbb{R} \times \mathbb{R}$ can be represented on the plane in such a way that there is a (1–1) correspondence between the elements of $\mathbb{R} \times \mathbb{R}$ and points of the plane.

The geometrical plane on which coordinate system has been specified is called the **real plane** or the **coordinate plane.**

Ordinarily we do not distinguish between the Cartesian plane $\mathbb{R} \times \mathbb{R}$ and the coordinate plane whose points correspond to or represent the elements of $\mathbb{R} \times \mathbb{R}$.

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If a point A of the coordinate plane corresponds to the ordered pair (a, b) then a, b are called the **coordinates** of A. a is called the x-coordinate or **abscissa** and b is called the y – coordinate or **ordinate.**

In the figure shown above, the coordinates of the points B,C,D and E are (3,2), (−4,3), (−3, −4) and (5, −4) respectively.

Corresponding to every ordered pair $(a, b) \in \mathcal{R}\times\mathcal{R}$ there is one and only one point in the plane and corresponding to every point in the plane there is one and only one ordered pair (a, b) in $\mathcal{R}\times\mathcal{R}$.

There is thus a (1–1) correspondence between $\mathcal{R}\times\mathcal{R}$ and the plane.

1.6 Geometrical Representation of Complex Numbers The Complex Plane

We have seen that there is a (1–1) correspondence between the elements (ordered pairs) of the Cartesian plane $\mathcal{R}\times\mathcal{R}$ and the complex numbers. Therefore, there is a (1–1) correspondence between the points of the coordinate plane and the complex numbers. We can, therefore, represent complex numbers by points of the coordinate plane. In this representation every complex number will be represented by one and only one point of the coordinate plane and every point of the plane will represent one and only one complex number. The components of the complex number will be the coordinates of the point representing it. In this representation the ***x*-axis** is called the **real axis** and the ***y*-axis** is called the **imaginary axis.** The coordinate plane itself is called the **complex plane** or **z – plane.**

By way of illustration a number of complex numbers have been shown in figure 3.

The figure representing one or more complex numbers on the *complex plane* is called an **Argand diagram.** Points on the ***x*-axis** represent **real numbers** whereas the points on the ***y*-axis** represent **imaginary numbers.**

In fig (4), x,y are the coordinates of a point.

It represents the complex number $x + iy$. The *real number*

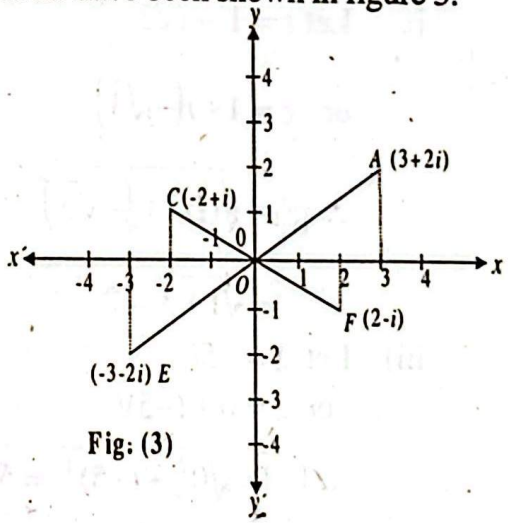


Fig: (3)

$\sqrt{x^2+y^2}$ **is called the modulus of the complex number** $a+ib$.

In the figure $\overline{MA} \perp \overrightarrow{o\ x}$

$\therefore\ \overline{OM} = x,\ \overline{MA} = y$

In the right-angled triangle OMA, we have, by Pythagoras theorem,

$$|\overline{OA}|^2 = |\overline{OM}|^2 + |\overline{MA}|^2$$

$$\therefore |\overline{OA}| = \sqrt{x^2+y^2}$$

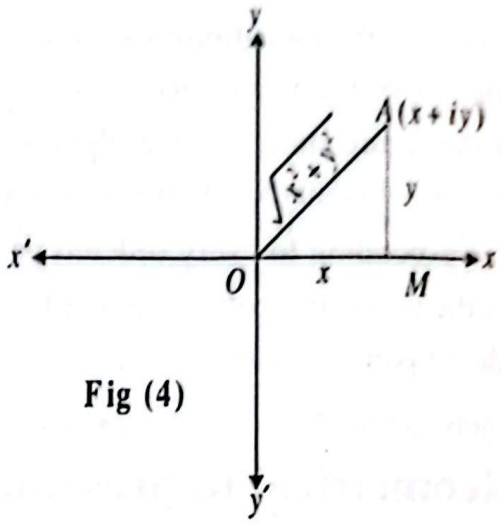


Fig (4)

Thus $|\overline{OA}|$ represents the modulus of $x + iy$. In other words: **The modulus of a complex number is the distance from the origin of the point representing the number.** The modulus of a complex number is generally denoted as: $|x+yi|$ or $|(x,y)|$. For convenience, a complex number is denoted by z.

If $z = x + iy = (x,y)$, then

$$|z| = \sqrt{x^2+y^2}$$

Example 1: Find moduli of the following complex numbers :

(i) $1-i\sqrt{3}$ (ii) 3 (iii) $-5i$ (iv) $3+4i$

Solution:

i) Let $z = 1 - i\sqrt{3}$

or $z = 1 + i\left(-\sqrt{3}\right)$

$$\therefore |z| = \sqrt{(1)^2 + \left(-\sqrt{3}\right)^2}$$

$$= \sqrt{1+3} = 2$$

ii) Let $z = 3$

or $z = 3 + 0.i$

$$\therefore |z| = \sqrt{(3)^2 + (0)^2} = 3$$

iii) Let $z = -5i$

or $z = 0 + (-5)i$

$$\therefore |z| = \sqrt{0^2 + (-5)^2} = 5$$

iv) Let $z = 3 + 4i$

$$\therefore |z| = \sqrt{(3)^2 + (4)^2}$$

$$= \sqrt{9+16} = 5$$

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Theorems: $\forall\ z, z_1, z_2 \in C$,

i) $|-z| = |z| = |\bar{z}| = |-\bar{z}|$ ii) $\bar{\bar{z}} = z$

iii) $z\bar{z} = |z|^2$ iv) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

v) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2},\ z_2 \neq 0$ vi) $|z_1 . z_2| = |z_1| . |z_2|$

vii) $|z_1| - |z_2| \leq |z_1 + z_2| \leq |z_1| + |z_2|$

Proof :(i): Let $z = a + ib$,

So, $-z = -a - ib, \bar{z} = a - ib$ and $-\bar{z} = -a + ib$

$$\therefore |-z| = \sqrt{(-a)^2 + (-b)^2} = \sqrt{a^2 + b^2} \quad (1)$$

$$|z| = \sqrt{a^2 + b^2} \quad (2)$$

$$|\bar{z}| = \sqrt{(a)^2 + (-b)^2} = \sqrt{a^2 + b^2} \quad (3)$$

$$|-\bar{z}| = \sqrt{(-a)^2 + (b)^2} = \sqrt{a^2 + b^2} \quad (4)$$

By equations (1), (2), (3) and (4) we conclude that

$$|-z| = |z| = |\bar{z}| = |-\bar{z}|$$

(ii) Let $z = a + ib$

so that $\bar{z} = a - ib$

Taking conjugate again of both sides, we have

$$\bar{\bar{z}} = a + ib = z$$

(iii) Let $z = a + ib$ so that $\bar{z} = a - ib$

$$\therefore z.\bar{z} = (a + ib)(a - ib)$$
$$= a^2 - iab + iab - i^2 b^2$$
$$= a^2 - (-1)b^2$$
$$= a^2 + b^2 = |z|^2$$

(iv) Let $z_1 = a + ib$ and $z_2 = c + id$, then

$$z_1+z_2=(a+ib)+(c+id)$$
$$=(a+c)+i(b+d)$$
$$\text{so, } \overline{z_1+z_2}=\overline{(a+c)+i(b+d)} \qquad \text{(Taking conjugate on both sides)}$$
$$=(a+c)-i(b+d)$$
$$=(a-ib)+(c-id)=\bar{z}_1+\bar{z}_2$$

(v) Let $z_1=a+ib$ and $z_2=c+id$, where $z_2\neq 0$, then

$$\frac{z_1}{z_2}=\frac{a+ib}{c+id}$$
$$=\frac{a+ib}{c+id}\times\frac{c-id}{c-id} \qquad \text{(Note this step)}$$
$$=\frac{(ac+bd)+i(bc-ad)}{c^2+d^2}=\frac{ac+bd}{c^2+d^2}+i\frac{bc-ad}{c^2+d^2}$$
$$\therefore \overline{\left(\frac{z_1}{z_2}\right)}=\overline{\frac{ac+bd}{c^2+d^2}+i\frac{bc-ad}{c^2+d^2}}$$
$$=\frac{ac+bd}{c^2+d^2}-i\frac{bc-ad}{c^2+d^2} \qquad (1)$$

Now
$$\frac{\bar{z}_1}{\bar{z}_2}=\frac{\overline{a+ib}}{\overline{c+id}}=\frac{a-ib}{c-id}$$
$$=\frac{a-ib}{c-id}\times\frac{c+id}{c+id}$$
$$=\frac{(ac+bd)-i(bc-ad)}{c^2+d^2}$$
$$=\frac{ac+bd}{c^2+d^2}-i\frac{bc-ad}{c^2+d^2} \qquad (2)$$

From (1) and (2), we have
$$\overline{\left(\frac{z_1}{z_2}\right)}=\frac{\bar{z}_1}{\bar{z}_2}$$

(vi) Let $z_1=a+ib$ and $z_2=c+id$, then

$$|z_1.z_2|=|(a+ib)(c+id)|$$
$$=|(ac-bd)+(ad+bc)i|$$
$$=\sqrt{(ac-bd)^2+(ad+bc)^2}$$
$$=\sqrt{a^2c^2+b^2d^2+a^2d^2+b^2c^2}$$
$$=\sqrt{(a^2+b^2)(c^2+d^2)}$$

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$= |z_1| . |z_2|$

This result may be stated thus: -

The modulus of the product of two complex numbers is equal to the product of their moduli.

(vii) Algebraic proof of this part is tedius. Therefore, we prove it geometrically.

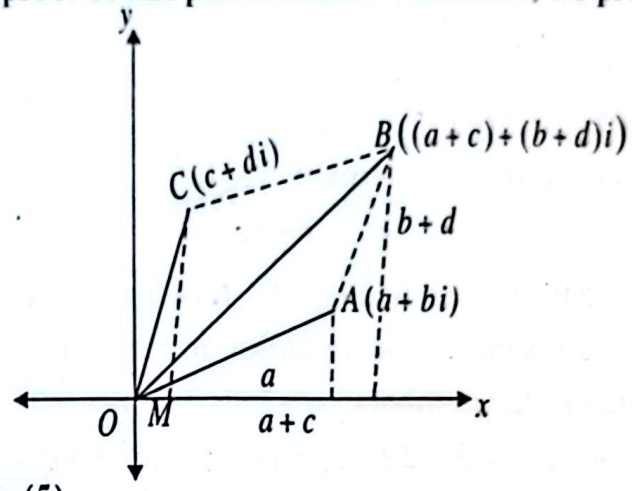


Fig (5)

In the figure point A represents $z_1 = a + ib$ and point C represents $z_2 = c + id$. We complete the parallelogram $OABC$. From the figure, it is evident that coordinates of B are $(a + c, b + d)$, therefore, B represents

$z_1 + z_2 = (a + c) + (b + d)\, i$ and $|\overline{OB}| = |z_1 + z_2|$.

Also $|\overline{OA}| = |z_1|$, $\quad |\overline{AB}| = |\overline{OC}| = |z_2|$.

In the ΔOAB; $OA + AB > OB$ $\quad (OA = m\overline{OA}$ etc.)

$\therefore |z_1| + |z_2| > |z_1 + z_2|$ (1)

Also in the same triangle, $OA - AB < OB$

$\therefore |z_1| - |z_2| < |z_1 + z_2|$ (2)

Combining (1) and (2), we have

$|z_1| - |z_2| < |z_1 + z_2| < |z_1| + |z_2|$ (3)

which gives the required results with inequality signs.

Results with equality signs will hold when the points A and C representing z_1 and z_2 become collinear with B. This will be so when $\frac{a}{b} = \frac{c}{d}$ (see fig (6)).

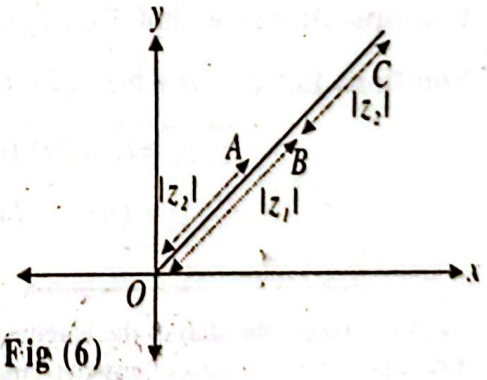


Fig (6)

In such a case

$$|z_1| + |z_2| = |\overline{OB}| + |\overline{OA}|$$
$$= |\overline{OB}| + |\overline{BC}|$$
$$= |\overline{OC}|$$
$$= |z_1 + z_2|$$

Thus $|z_1 + z_2| = |z_1| + |z_2|$

The second part of result (vii) namely

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

is analogue of the triangular inequality*. In words, it may be stated thus: - The modulus of the sum of two complex numbers is less than or equal to the sum of the moduli of the numbers.

Example 2: If $z_1 = 2 + i$, $z_2 = 3 - 2i$, $z_3 = 1 + 3i$ then express

$\dfrac{\overline{z_1}\,\overline{z_3}}{z_2}$ in the form $a + ib$

(Conjugate of a complex number z is denoted as $\overline{z}$)

Solution:

$$\frac{\overline{z_1}\,\overline{z_3}}{z_2} = \frac{(\overline{2+i})(\overline{1+3i})}{3-2i} = \frac{(2-i)\ (1-3i)}{3-2i}$$
$$= \frac{(2-3)+(-6-1)i}{3-2i} \qquad = \frac{-1-7i}{3-2i}$$
$$= \frac{(-1-7i)\ (3+2i)}{(3-2i)\ (3+2i)}$$
$$= \frac{(-3+14)+(-2-21)i}{3^2+2^2} \quad = \frac{11}{13} - \frac{23}{13}i$$

Example 3: Show that $\forall z_1, z_2 \in C$, $\overline{z_1\ z_2} = \overline{z_1}\ \overline{z_2}$

Solution: Let $z_1 = a + bi$, $z_2 = c + di$

$$\overline{z_1\ z_2} = \overline{(a+bi)\ (c+di)} = \overline{(ac-bd)+(ad+bc)i}$$
$$= (ac-bd) - (ad+bc)\,i \qquad (1)$$

---

* In any triangle the sum of the lengths of any two sides is greater than the length of the third side and difference of the lengths of any two sides is less than the length of the third side.

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$$\overline{z_1}\cdot\overline{z_2} = \overline{(a+bi)}\ \overline{(c+di)}$$
$$= (a-bi)\ (c-di)$$
$$= (ac-bd) + (-ad-bc)i \qquad (2)$$

Thus from (1) and (2) we have, $\overline{z_1\ z_2} = \overline{z_1}\ \overline{z_2}$

Polar form of a Complex number : Consider adjoining diagram representing the complex number $z = x+iy$. From the diagram, we see that $x = r\cos\theta$ and $y = r\sin\theta$ where $r = |z|$ and θ is called argument of z.

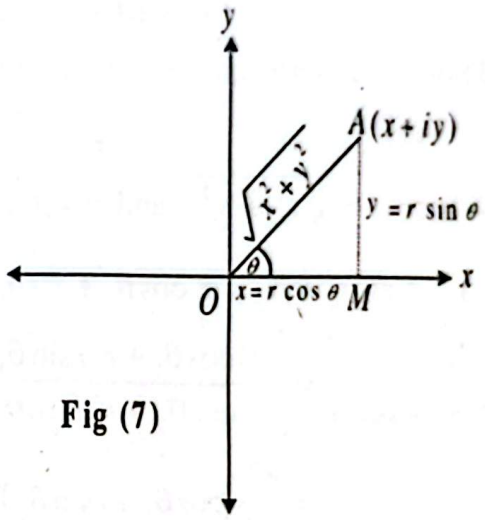


Fig (7)

Hence $x+iy = r\cos\theta + ir\sin\theta$ ....(i)

where $r = \sqrt{x^2+y^2}$ and $\theta = \tan^{-1}\frac{y}{x}$

Equation (i) is called the polar form of the complex number z.

Example 4: Express the complex number $1+i\sqrt{3}$ in polar form.

Solutoin :

Step-I : Put $r\cos\theta = 1$ and $r\sin\theta = \sqrt{3}$

Step-II : $r^2 = (1)^2 + \left(\sqrt{3}\right)^2$

$\Rightarrow r^2 = 1+3 = 4 \quad \Rightarrow r = 2$

Step-III : $\theta = \tan^{-1}\frac{\sqrt{3}}{1} = \tan^{-1}\sqrt{3} = 60°$

Thus $1+i\sqrt{3} = 2\cos 60° + i\,2\sin 60°$

De Moivre's Theorem : -

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta, \forall n \in Z$$

Proof of this theorem is beyond the scope of this book.

1.7 To find real and imaginary parts of

i) $(x+iy)^n$ ii) $\left(\frac{x_1+iy_1}{x_2+iy_2}\right)^n$, $x_2+iy_2 \neq 0$

for $n = \pm1, \pm2, \pm3, \ldots$

i) Let $x = r\cos\theta$ and $y = r\sin\theta$, then

$(x+iy)^n = (r\cos\theta + ir\sin\theta)^n$

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$$= (r\cos\theta + ir\sin\theta)^n$$
$$= [r(\cos\theta + i\sin\theta)]^n$$
$$= r^n(\cos\theta + i\sin\theta)^n$$
$$= r^n(\cos n\theta + i\sin n\theta) \qquad \text{(By De Moivre's Theorem)}$$
$$= r^n\cos n\theta + ir^n\sin n\theta$$

Thus $r^n\cos n\theta$ and $r^n\sin n\theta$ are respectively the real and imaginary parts of $(x+iy)^n$.

Where $r = \sqrt{x^2+y^2}$ and $\theta = \tan^{-1}\frac{y}{x}$.

ii) Let $x_1 + iy_1 = r_1\cos\theta_1 + r_1\sin\theta_1$ and $x_2 + iy_2 = r_2\cos\theta_2 + r_2\sin\theta_2$ then,

$$\left(\frac{x_1+iy_1}{x_2+iy_2}\right)^n = \left(\frac{r_1\cos\theta_1 + r_1 i\sin\theta_1}{r_2\cos\theta_2 + r_2 i\sin\theta_2}\right)^n = \frac{r_1{}^n(\cos\theta_1 + i\sin\theta_1)^n}{r_2{}^n(\cos\theta_2 + i\sin\theta_2)^n}$$

$$= \frac{r_1^n}{r_2^n}(\cos\theta_1 + i\sin\theta_1)^n(\cos\theta_2 + i\sin\theta_2)^{-n}$$

$$= \frac{r_1^n}{r_2^n}(\cos n\theta_1 + i\sin n\theta_1)(\cos(-n\theta)_2 + i\sin(-n\theta_2)),$$

(By De Moivre's Theorem

$$= \frac{r_1^n}{r_2^n}(\cos n\theta_1 + i\sin n\theta_1)(\cos n\theta_2 - i\sin n\theta_2), \quad (\cos(-\theta) = \cos\theta$$

$$\sin(-\theta) = -\sin\theta)$$

$$= \frac{r_1^n}{r_2^n}, [(\cos n\theta_1\cos n\theta_2 + \sin n\theta_1\sin n\theta_2)$$
$$+ i(\sin n\theta_1\cos n\theta_2 - \cos n\theta_1\sin n\theta_2)]$$

$$= \frac{r_1^n}{r_2^n}[\cos(n\theta_1 - n\theta_2) + i\sin(n\theta_1 - n\theta_2)] \quad \because \cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\si$$

and $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin$

$$= \frac{r_1^n}{r_2^n}[\cos n(\theta_1 - \theta_2) + i\sin n(\theta_1 - \theta_2)]$$

$$= \frac{r_1^n}{r_2^n}[\cos n(\theta_1 - \theta_2) + i\sin n(\theta_1 - \theta_2)]$$

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Thus $\frac{r_1^n}{r_2^n}\cos n(\theta_1-\theta_2)$ and $\frac{r_1^n}{r_2^n}\sin n(\theta_1-\theta_2)$ are respectively the real and imaginary parts of $\left(\frac{x_1+iy_1}{x_2+iy_2}\right)^n$, $x_2+iy_2\neq 0$

where $r_1=\sqrt{x_1^2+y_1^2}$; $\theta_1=\tan^{-1}\frac{y_1}{x_1}$ and $r_2=\sqrt{x_2^2+y_2^2}$; $\theta_2=\tan^{-1}\frac{y_2}{x_2}$

Example 5: Find out real and imaginary parts of each of the following complex numbers.

i) $(\sqrt{3}+i)^3$ ii) $\left(\frac{1-\sqrt{3}\,i}{1+\sqrt{3}\,i}\right)^5$

Solution :

i) Let $r\cos\theta=\sqrt{3}$ and $r\sin\theta=1$ where

$r^2=(\sqrt{3})^2+1^2$ or $r=\sqrt{3+1}=2$ and $\theta=\tan^{-1}\frac{1}{\sqrt{3}}=30^\circ$

So, $(\sqrt{3}+i)^3=(r\cos\theta+ir\sin\theta)^3$

$=r^3(\cos 3\theta+i\sin 3\theta)$ (By De Moivre's Theorem)

$=2^3(\cos 90^\circ+i\sin 90^\circ)$

$=8(0+i.1)$

$=8i$

Thus 0 and 8 are respectively real and imaginary Parts of $(\sqrt{3}+i)^3$.

ii) Let $r_1\cos\theta_1=1$ and $r_1\sin\theta_1=-\sqrt{3}$

$\Rightarrow r_1=\sqrt{(1)^2+(-\sqrt{3})^2}=\sqrt{1+3}=2$ and $\theta_1=\tan^{-1}-\frac{\sqrt{3}}{1}=-60^\circ$

Also let $r_2\cos\theta_2=1$ and $r_2\sin\theta_2=\sqrt{3}$

$\Rightarrow r_2=\sqrt{(1)^2+(\sqrt{3})^2}=\sqrt{1+3}=2$ and $\theta_2=\tan^{-1}\frac{\sqrt{3}}{1}=60^\circ$

So $\left(\frac{1-\sqrt{3}\,i}{1+\sqrt{3}\,i}\right)^5=\left[\frac{2(\cos(-60^\circ)+i\sin(-60^\circ))}{2(\cos(60^\circ)+i\sin(60^\circ))}\right]^5$

$=\frac{(\cos(-60^\circ)+i\sin(-60^\circ))^5}{(\cos(60^\circ)+i\sin(60^\circ))^5}$

$=(\cos(-60^\circ)+i\sin(-60^\circ))^5(\cos(60^\circ)+i\sin(60^\circ))^{-5}$

$= (\cos(-300^\circ) + i\sin(-300^\circ))(\cos(-300^\circ) + i\sin(-300^\circ))$ By De Moivre's Theorem

$= (\cos(300^\circ) - i\sin(300^\circ))(\cos(300^\circ) - i\sin(300^\circ))$ $\because \cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$

$$= \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2 = \frac{-1}{2} + \frac{\sqrt{3}}{2}i$$

Thus $\frac{-1}{2}, \frac{\sqrt{3}}{2}$ are respectively real and imaginary parts of $\left(\frac{1-\sqrt{3}\,i}{1+\sqrt{3}\,i}\right)^5$

Exercise 1.3

1. Graph the following numbers on the complex plane: -

i) $2+3i$ ii) $2-3i$ iii) $-2-3i$ iv) $-2+3i$

v) -6 vi) i vii) $\frac{3}{5} - \frac{4}{5}i$ viii) $-5-6i$

2. Find the multiplicative inverse of each of the following numbers: -

i) $-3i$ ii) $1-2i$ iii) $-3-5i$ iv) $(1,2)$

3. Simplify

i) i^{101} ii) $(-ai)^4, a \in \mathbb{R}$ iii) i^{-3} iv) i^{-10}

4. Prove that $\bar{z} = z$ iff z is real.

5. Simplify by expressing in the form $a+bi$

i) $5+2\sqrt{-4}$ ii) $(2+\sqrt{-3})(3+\sqrt{-3})$

iii) $\frac{2}{\sqrt{5}+\sqrt{-8}}$ iv) $\frac{3}{\sqrt{6}-\sqrt{-12}}$

6. Show that $\forall\, z \in C$

i) $z^2 + \bar{z}^2$ is a real number. ii) $(z-\bar{z})^2$ is a real number.

7. Simplify the following

i) $\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^3$ ii) $\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^3$

iii) $\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^{-2}\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$ iv) $(a+bi)^2$

v) $(a+bi)^{-2}$ vi) $(a+bi)^3$

vii) $(a-bi)^3$ viii) $(3-\sqrt{-4})^{-3}$

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